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XII-P4
Kavya.S

QMD

12 . Std

Quarterly Examination - 2026
PHYSICS

1	2	2	1	7	

Time : 3 00 Hrs

Marks : 70

PART- I

15 x 1 = 15

Note : ***(i) Answer all the questions.***

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1. The heat energy produced in a resistance of 10 ohm when 5A current flows through it for 5 minutes
(a) 75kJ (b) 75J (c) 2500 J (d) 2500 kJ

2. Which charge configuration produces a uniform electric field?
(a) point charge (b) uniformly charged infinite line
(c) uniformly charged infinite plane (d) uniformly charged spherical shell

3. A Parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?
(a) capacitance (b) charge (c) voltage (d) energy density

4. A carbon resistor of (47 ± 4.7) k Ω to be marked with rings of different colours for its identification. The colour code sequence will be
(a) Yellow – Green – Violet – Gold (b) Yellow – Violet – Orange – Silver
(c) Violet – Yellow – Orange – Silver (d) Green – Orange – Violet - Gold

5. The temperature coefficient of resistance of a wire is 0.00125 per °C. At 20°C, its resistance is 1 Ω. The resistance of the wire will be 2 Ω at
(a) 800° C (b) 700° C (c) 820° C (d) 820° C

6. A particle having mass m and charge q accelerated through a potential difference V. Find the force experienced when it is kept under perpendicular magnetic field B
(a) $\sqrt{\frac{2q^3B^2V}{m}}$ (b) $\sqrt{\frac{q^3B^2V}{2m}}$ (c) $\sqrt{\frac{2q^3B^2V}{m}}$ (d) $\sqrt{\frac{2q^3BV}{m^3}}$

7. A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about its axle with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is
(a) $\frac{q}{m}$ (b) $\frac{2q}{m}$ (c) $\frac{q}{2m}$ (d) $\frac{q}{4m}$

8. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is
(a) 2 A (b) 18 A (c) 12 A (d) 1 A

9. In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
(a) $\frac{Q}{2}$ (b) $\frac{Q}{\sqrt{5}}$ (c) $\frac{Q}{\sqrt{2}}$ (d) Q

10. Fraunhofer lines are an example of _____ spectrum.
(a) line erhiesion (b) line absorption (c) band emission (d) band absorption

11. The electric and magnetic fields of an electromagnetic wave are
(a) in phase and perpendicular to each other
(b) out of phase and not perpendicular to each other
(c) in phase and not perpendicular to each other
(d) out of phase and perpendicular to each other

12. For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is. (a) 30° (b) 45° (c) 60° (d) 90°

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13 Stars twinkle due to,
(a) reflection (b) total internal reflection (c) refraction (d) polarization

14. The number of electrons in one coloumb charge
(a) 1.6×10^{-19} (b) 1.6×10^{19} (c) 6.25×10^{18} (d) 6.25×10^{-18}

15. The Ratio of the Magnetic length and its geometrical length is
(a) 5/8 (b) 0 893 (c) both (a) and (b) (d) none of the above

PART- II

Answer any six questions and Question No.23 is compulsory. 6 x 2 = 12

16. Why is it safer to be inside a car than standing under a tree during lightning?
17. Distinguish drift velocity and mobility
18. State Lenz's law.
19. How the current sensitivity of a galvanometer can be increased?
20. How will you define Q-factor?
21. Give any two uses of Microwaves.
22. A copper wire of cross-sectional area 0 5mnf carries a current of 0 2A. If the free electron density of copper is $8.4 \times 10^{28}\ m^{-3}$, then compute the drift velocity of electrons.
23. The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity of the medium is 2.25. Compute the refractive index of the medium.
24. Write the two conditions for total internal reflection.

PART- III

Answer any six questions and Question No. 33 is compulsory. 6 x 3 = 18

25. Derive an expression for energy stored in capacitor?
26. State the applications of Seebeck effect.
27. How is a galvanometer converted into ammeter?
28. How will you induce an emf by changing the area enclosed by the coil?
29. Write down the six properties of electromagnetic waves?
30. Derive the relation between f and R for a spherical mirror?
31. Obtain Gauss's law from Coulomb's law.
32. Find the impedance of a series RLC circuit if the inductive reactance, capacitive reactance and resistance are 184 Ω, 144 Ω and 30 Ω respectively. Also calculate the phase angle between voltage and current.
33. State Kirchhoff's junction rule and loop rule.

PART - IV

5 x 5 = 25

Answer all Questions :

34 A) Obtain an expression for electric field due to an infinitely long charged wire. *(OR)*
B) (I) Distinguish between Coulomb force and Gravitational force. (3 mark)
(II) Calculate the electric flux through the rectangle of sides 5 cm and 10 cm kept at angle of 60° in the region of a uniform electric field 100 NC^{-1} (2 mark)

35. A) How the emf of two cells are compared using potentiometer? *(OR)*
B) Describe the microscopic model of current and obtain general form of ohm's law.

36. A) (I) Explain Lorentz force. (3 mark)
(II) Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1 A at distance of 1m from it. (2 mark) *(OR)*
B) Obtain lens maker's formula

37 A) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit. *(OR)*
B) Explain the construction and working of a transformer

38. A) Explain the types of emission spectrum. *(OR)*
B) Explain the Maxwell's modification of Ampere's circuital law.

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